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SUBMITTED ELECTRONICALLY VIA www.regulations.gov

Dear Ms. Carlton and Dr. Posnack,

Noblis, an independent nonprofit organization chartered to work in the public interest, appreciates the opportunity to comment on the Centers for Medicare and Medicaid Services (CMS)' Request for Information (RFI) on the Health Technology Ecosystem, 2025-08701 *(90 Fed. Reg. 21034).*

Established in 1996, Noblis is a nonprofit 501(c)(3) science, technology, and strategy organization. Our team of objective and independent engineers, solution developers and subject matter experts serve the public good, enabling Noblis to work on the government's side of the table to solve complex problems of national importance. Noblis applies the best technical thought and engineering know-how and finds solutions that are practical, efficient, and applicable to agency priorities. With more than 2,100 technical and support staff, Noblis has supported more than 65 federal agencies, including most cabinet-level departments. We work in a wide range of industry sectors such as defense, homeland security, intelligence, law enforcement, and Federal civilian sectors, including over 25 years working with the Centers for Medicare and Medicaid Services.

Noblis invests in our people and innovations through our internally-funded Noblis Sponsored Research (NSR) program including ten lab environments that accelerate short-turnaround research projects while facilitating the development of long-term innovations covering artificial intelligence/machine learning (AI/ML), advanced analytics, cybersecurity, virtual/augmented reality (VR/AR), high-performance computing (HPC), digital engineering, orchestrated autonomy, applied science, and digital transformation. We are problem-solvers and doers with an accelerated prototype-to-mission mindset and are excited to be part of the comment period and look forward to working with CMS and ASTP/ONC on the Health Technology Ecosystem Request for Information, 2025-08701 (*90 Fed. Reg. 21034*).

Thank you for the opportunity to submit these comments.

Sincerely,

John Beresny

John Beresny

Comment Response

Below, we have categorically outlined our supporting comments by the original questions posed by CMS.

Patient Needs

PC-1. What health management or care navigation apps would help you understand and manage your (or your loved ones') health needs, as well as the actions you should take?

Digital health-forward care navigation apps should make it effortless for patients, caregivers, and clinicians to share data, coordinate services, and act on clear, personalized insights. To do so, apps must (1) adopt common interoperability standards (HL7 FHIR, DICOM, LOINC, ICD-11, SNOMED CT) so information flows wherever care happens, and (2) pair that technical backbone with human-centered design (HCD) that puts the patient experience first—from the moment an appointment is booked to long-term condition management.

- **Friction-free scheduling** – One interface to book, reschedule, or cancel visits without phone calls or multiple log-ins.
- **Cross-provider data exchange** – Real-time interoperability (e.g., via Mirth Connect, Redox, Google Cloud Healthcare Application Programming Interface [API]) that links clinics, labs, transport, and tele-triage in a single workflow.
- **Personalized, data-driven guidance** – Standards-based integrations that turn real-time vitals and records into concrete nutrition, activity, and medication prompts.
- **Patient-friendly AI touchpoints** – Passive/active AI delivered through channels users already trust (secure text, voice assistants) to refill meds or message clinicians without extra apps.
- **Blockchain-grade privacy** – Decentralized, immutable ledgers that give patients key-based control over who sees what, slashing breach risk and admin overhead.
- **Unified health dashboards** – Consumer-grade hubs (think Apple Health-style) that aggregate wearables, Electronic Health Records (EHR) data, and physician-ordered tests into shareable, easy-to-read trends.
- **Insight-centric trackers** – Apps like Levels, Gentler Streak, and Ada that translate glucose, activity, and symptom data into plain-language cause-and-effect lessons users can act on.
- **AI care advocates** – Conversational agents that reframe medical jargon, surface "why" behind decisions, and bridge language or literacy gaps for patients and families.

PC-5. What can CMS and its partners do to encourage patient and caregiver interest in these digital health products?

CMS can wield its purchasing power to accelerate digital health adoption—both by rewarding data-sharing and by hard-wiring patient-centered metrics into every contract. Doing so will not only modernize Medicare, Medicaid, and the Children's Health Insurance Program (CHIP) but also push the wider market toward integrated, value-based care.

- **Open-data Medicare contracts** – Require every Medicare Administrative Contractor (MAC) to make claims and encounter data openly available to providers and payers via standardized APIs.
- **Mandatory data exchange conditions** – Make EHR sharing with patients, Qualified Health Information Networks (QHINs), and Health Information Exchanges (HIEs) a new condition of Medicare participation, separate from payment rules.
- **Patient-centric quality metrics** – Adopt frameworks such as the National Academy of Medicine's to track beneficiary sentiment, access, and service satisfaction inside value-based payment models.
- **ICD-11 modernization** – Lead the U.S. transition to the fully digital ICD-11 platform so codes align with today's science, link seamlessly to SNOMED CT, and feed richer analytics for care improvement and research.
- **Exploding self-tracked data** – With 53% of U.S. adults wearing at least one connected device and 54% digitally tracking a health metric, consumer data now rivals EHR volume and must be harmonized[1].
- **Creating opportunities to incentivize patients and caregivers** – Enhance Accountable Care Organization (ACO) financial rewards by focusing on incentives that promote health data cross-sharing between provider groups and their patients through innovations such as health data from wearables and health improvement programs such as rewards for patients adhering to episodic immunization.

Data Access and Integration

PC-8c. What specific opportunities and challenges exist to improve accessibility, interoperability and integration of clinical data from different sources to enable more meaningful clinical research and generation of actionable evidence?

Healthcare data still lives in too many walled-off systems, each with its own schema and coding rules making large-scale analytics slow and costly. Modern ontology-discovery and mapping tools can break those walls down by creating shared vocabularies and data models on the fly, letting clinicians and analysts focus on insights instead of integration.

- **On-demand ontologies and taxonomies** – Auto-generate flexible, extensible data models that align granular elements to the right medical specialties, then splice in terms from related ontologies for quick cross-walks.
- **ML-driven concept fusion** – Combine rule-based mapping, machine learning, and large language models (LLMs) to spot equivalent codes across sources, slash manual work, and surface novel analytic groupings.

[1] From *Screenagers to Silver Surfers: How each generation clicks with care,* by Rock Health, 2025 (https://rockhealth.com/insights/screenagers-to-silver-surfers-how-each-generation-clicks-with-care/).

Ecosystem

TD-1. What short term (in the next 2 years) and longer-term steps can CMS take to stimulate developer interest in building digital health products for Medicare beneficiaries and caregivers?

An immediate jolt of "developer-friendliness" plus a clear runway for scaling can draw new talent and capital into solutions that serve Medicare beneficiaries and their caregivers. Below are near-term actions CMS can execute within two years, followed by longer-horizon moves that lock in momentum and create durable market signals.

Short-term (next 24 months)

- **Launch a Super-charged Blue Button Sandbox**
  Ship synthetic FHIR R4 datasets, consent-flow examples, and a command-line test harness so teams can prototype integrations in hours.
- **Run Problem-Backlog Micro-Grants and Hackathons**
  Post tightly scoped challenges (e.g., fall prevention, medication adherence) with prizes and offer live API support at annual FHIR hackathons.
- **Introduce Provisional CPT/G-codes for Digital Pilots**
  Allow billing for app-enabled monitoring or coaching during limited trials, giving startups a revenue path while CMS gathers evidence.
- **Publish an Open-Source Reference Stack**
  Host sample apps, SDKs, and unit tests in a CMS GitHub org, all mapped to certification criteria to slash redundant dev work.
- **Offer Value-Based Care App Bonuses**
  Provide small upward adjustments to bundled or shared-savings payments when providers prescribe CMS-certified apps and submit outcomes.

Longer-term (3–5 years and beyond)

- **Maintain a De-identified Outcomes Registry**
  Create a "data corridor" where developers can study longitudinal effectiveness without new IRB hurdles, accelerating iteration and evidence generation.
- **Elevate Beneficiary Portals to a Unified "Health Home"**
  Fund User Experience/User Interface (UX/UI) revamps and cross-provider data unification, making portals the single front door for scheduling, records, and caregiver access.
- **Expand AI Research and Deployment Grants**
  Focus on practical wins—diagnostic support, prior-auth automation, personalized med regimens—and embed voice/chat agents where seniors already engage.

- **Pioneer Smart Tele-Health Interfaces**
  Partner with consumer-electronics firms to bring tele-visits, med reminders, and appointment alerts to the living-room screens familiar to older adults.
- **Institutionalize Open-Source First**
  Reduce licensing costs, avoid vendor lock-in, and cultivate a public developer community by mandating Open-Source Software (OSS) components wherever privacy and security allow.

These parallel tracks (quick wins that remove friction today and structural moves that reward innovation tomorrow) can transform CMS from a payer into the nation's most powerful catalyst for age-friendly digital health.

In Conclusion

Noblis appreciates the opportunity to help shape CMS and ONC's vision for a patient-centered, data-driven health-technology ecosystem. Drawing on three decades of public-interest research and deep, hands-on work with CMS, we have outlined practical, near-term actions—super-charged developer sandboxes, open-data contracts, provisional billing codes—as well as longer-horizon investments in unified beneficiary portals, AI-enabled care, and open-source infrastructure. Together, these steps can accelerate market innovation, reduce administrative waste, and deliver measurable value to Medicare, Medicaid, and CHIP beneficiaries and their caregivers.

Noblis agrees with all the recommendations herein this reply and many recommendations made by Leavitt Partners, an HMA Company, in their white paper, *"Kill the Clipboard! A Federal Policy and Industry Roadmap to Accelerate Innovation and Cut Administrative Waste."*[2] We stand ready to collaborate with CMS, ONC, and industry stakeholders to translate these ideas into scalable solutions that improve outcomes and lower costs.

Thank you for considering our comments.

[2]From *Kill the Clipboard! A Federal Policy and Industry Roadmap to Accelerate Innovation and Cut Administrative Waste,* by Leavitt Partners, 2025 (https://leavittpartners.com/wp-content/uploads/2025/03/NEW-Ideas-for-the-Trump-Department-of-HHS_FINAL.pdf).